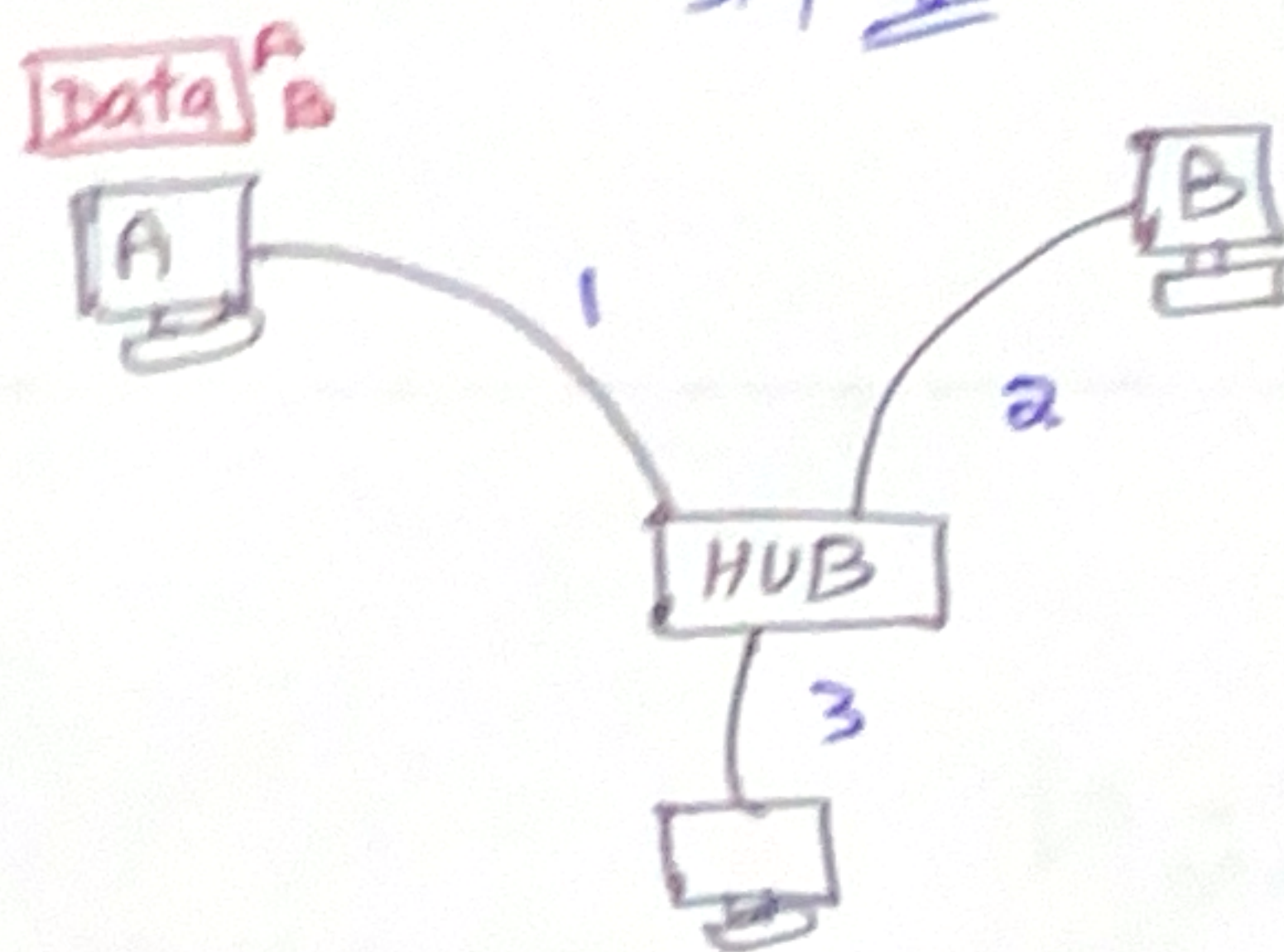


NETWORK DEVICE TEST

1) Explain behaviour of hub device?

- * It is used to connect the multiple network device.
- * It is a broadcast device.

Step by Step



* we have 3 pc's, name as A, B & C and port no is A-1, B-2, C-3.

* First of all, the data is send to PC A to PC B.

* First you have to send the data to hub, once the hub is received the data.

* the data is broadcast to all PC.

* Send PC B and PC C

* PC B receives the data and PC C receive and automatically drop the data.

* main thing, once the data is shared to hub, it can be broadcast and collision.

* Bcoz, of data loss and heavy traffic.

* It is called as half duplex.

* The hub behaviour is once you share the data from one pc to another pc, using the hub, the data can be broadcasted.

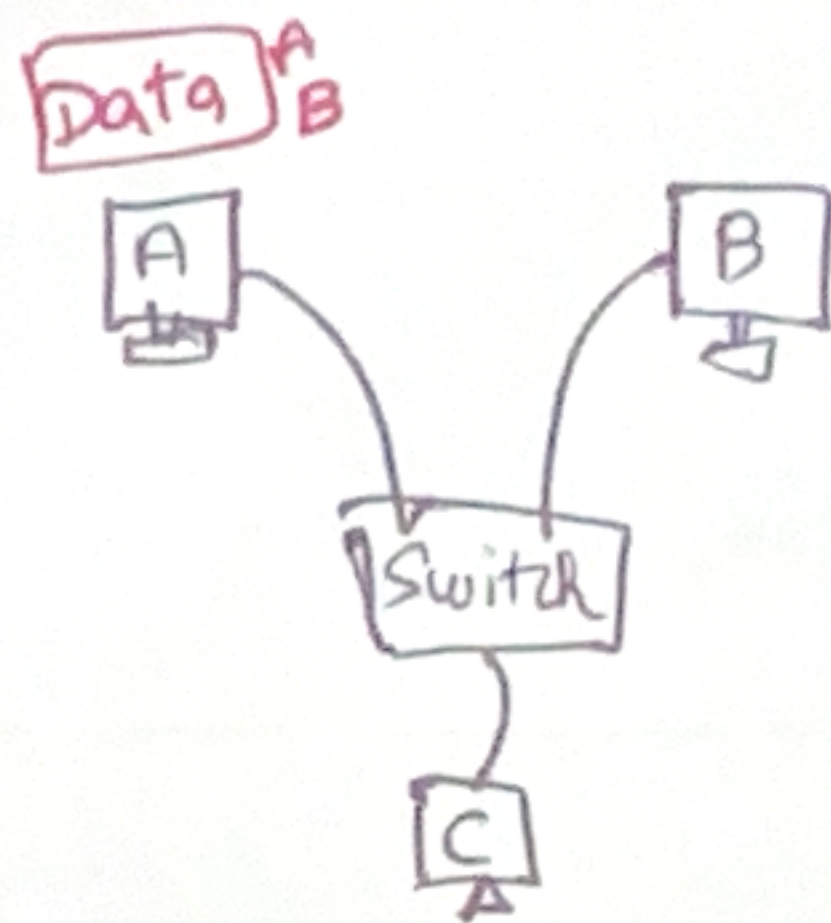
2) Explain behaviour of switch device?

Switch

It is used to connect multiple network device.

It contains mac table.

It will act as broadcast & unicast.



* Shared the data from one pc to another pc.

we have 3 pc = A, B, C.

* First you have to assign IP of A and B PC.

A = 50.0.0.10 & B = 50.0.0.20

we have to send the echo data from PC B with the help of Switch.

* Switch behaves like once data is received, it note the port number & mac address on the mac table is known as **Address learning**.

* First time, you will send the data and stores on the mac table and broadcast to all pc.

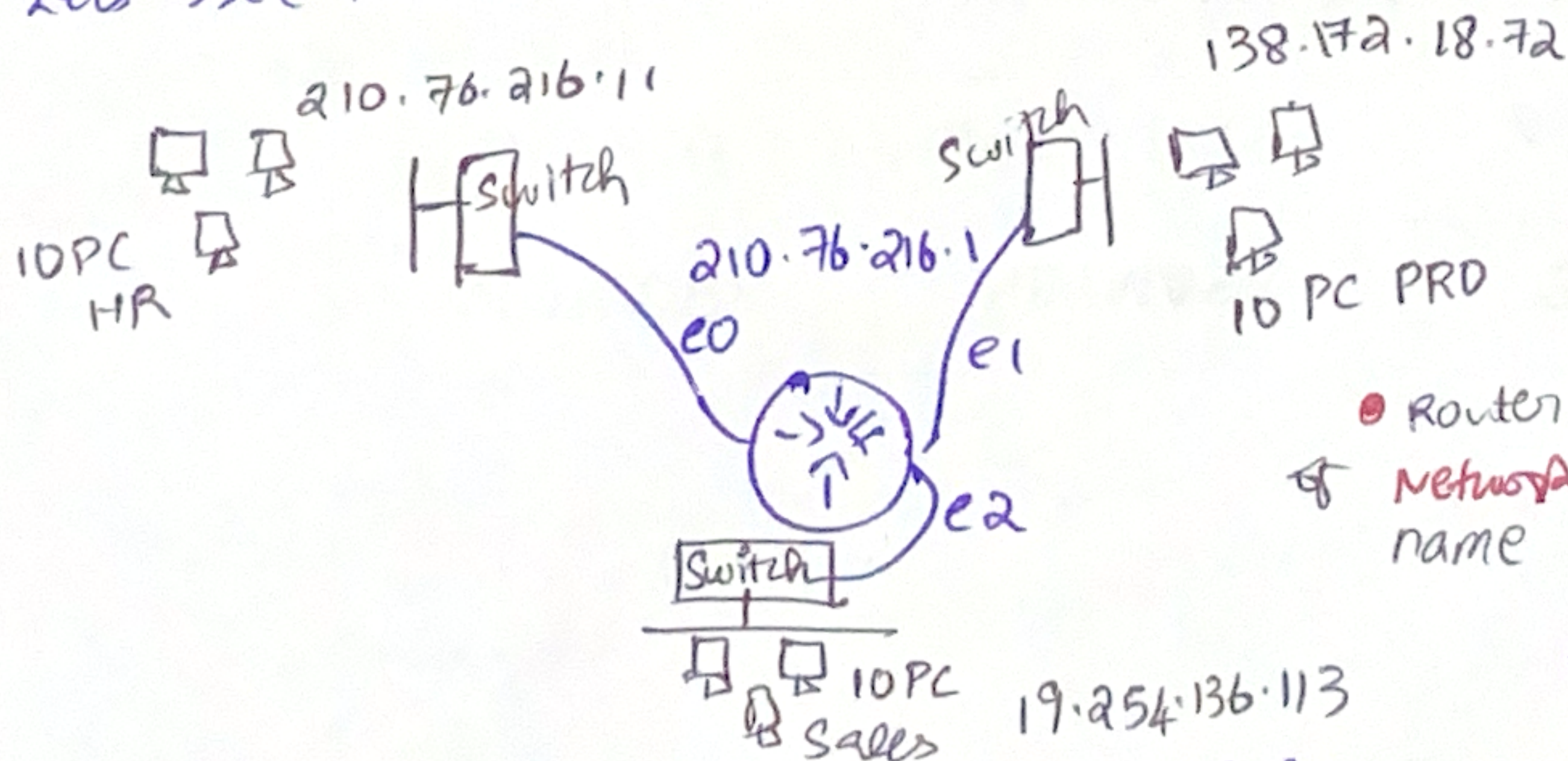
* Then return from PC B to PC A, it will comes in switch. It will noted PC B port number and mac address & check whether the destination is on the mac table and directly send to the destination PC.

* This will called as **broadcast + unicast**.

3) Explain behaviour of router device?

- * It use to breaks the broadcast domain.
- * It is a unicast device. Contains routing table.
- * Routing table contains list of rule.

Lets see the below diag:-



• Router device perform data sharing by the way of unicast.

• Each dept have a IP address and have assigned IP to router.

• Router device notes the information of **Network ID, prefix & Interface** name in the routing table.

Router → IP → 3 Information.

• we have 3 dept as HR, production & Sales.

• Each dept PC has connected by Switch and all the switches are connected by router device.